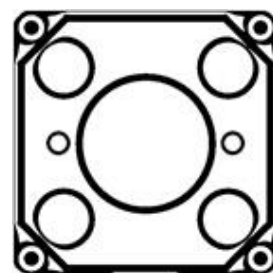




ifm_O2D5_O3D3_Vision Add-On Instruction Guide for Allen Bradley Compact/ControlLogix PLC's

O3D3xx Sensor





Version 1.0 1/2024



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1 Preliminary notes

This document demonstrates:

1. Ethernet/IP communication between an O3D3xx sensor and an Allen Bradley controller.
2. Functions of the ifm_O2D5_O3D3_Vision Add-On Instruction.

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2 Intended Use

The ifm_O2D5_O3D3_Vision Add-On Instruction provides a logical user interface to the O3D3xx sensor.

The Add-On Instruction is to be integrated in a PLC program to:

1. Provide evaluation results
2. Switch applications
3. Display application information
4. Request pass/fail statistics
5. Process errors

Logical integration of the Add-On Instruction is aided by its output tags. The outputs tags Camera Ready, Command Busy and Command Done provide the logic states of the O3D3xx sensor so the user controlled functions can be utilized with minimal processing outside the Add-On Instruction.

3 Prerequisites

The user has:

1. Assigned an IP address for the O3D3xx sensor.
2. Configured the O3D3xx sensor with one or more applications using the Vision Assistant configuration software.



4 Set O3D3xx to Ethernet/IP communication

1. In the Vision Assistant software, connect to the sensor and navigate to **Device setup**.
2. Under Device setup click **Interfaces**.
3. Set the Active fieldbus to **EtherNet/IP**.
4. The ifm_O2D5_O3D3_Vision Add-On Instruction was developed with the default **Producing size** of 450 bytes and **Consuming size** of 8 bytes. It is not recommended to change these settings.
5. Click the **save icon** to set the settings on the O3D3xx.

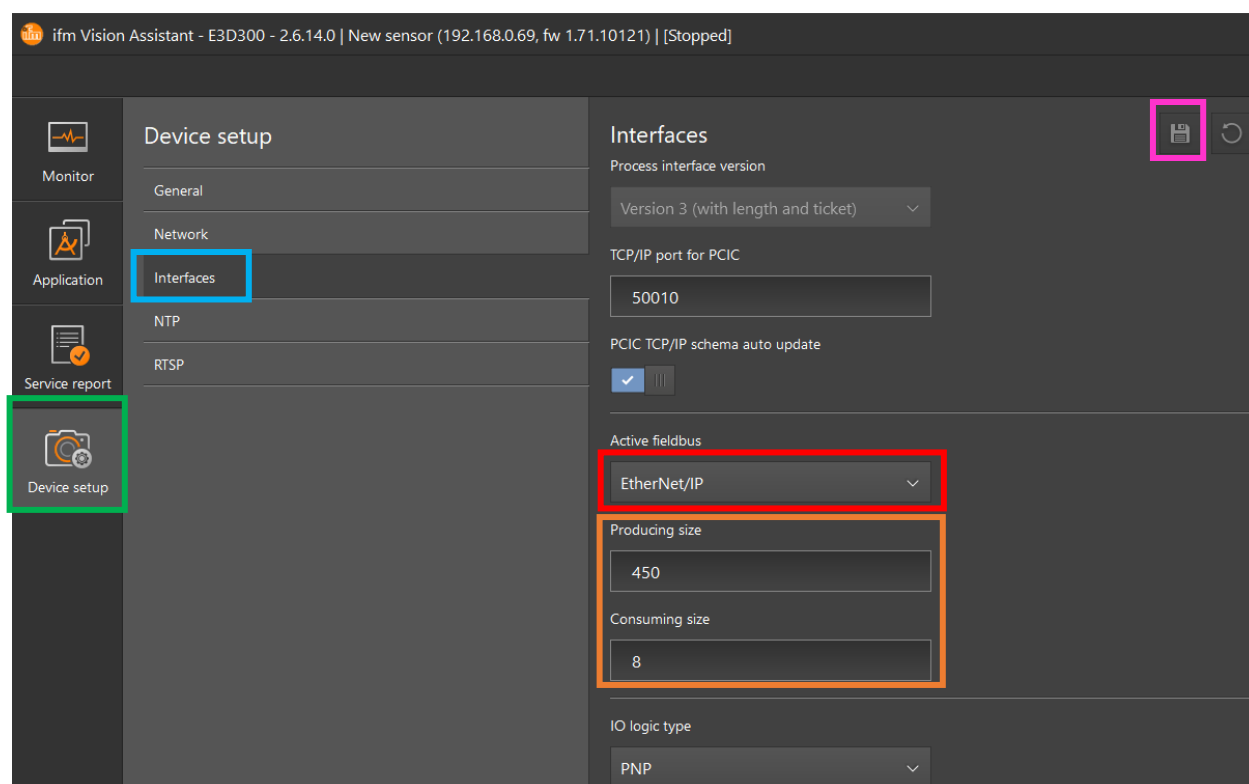


Figure 1



5 PLC setup

5.1 EDS file

There are two methods to find the O3D3xx EDS file for integration into the PLC. Use one of the methods in section 5.1 to save the EDS file to a PC.

5.1.1 EDS file from the O3D3xx_EIP_AllenBradley_StartupPackage

1. Find the **EDS file** for the O3D3xx in the folder named “**1. EDS**” in the O3D3xx_EIP_AllenBradley_StartupPackage.

Name	Date modified	Type	Size
1. EDS	11/15/2023 2:22 PM	File folder	
2. Add_On Instruction	11/15/2023 2:23 PM	File folder	
3. Sample Project	11/15/2023 12:10 PM	File folder	
Change Log.txt	11/15/2023 11:47 AM	Text Document	1 KB

Figure 2

Name	Date modified	Type	Size
O3D3-EDS-V2.2.eds	11/15/2023 2:22 PM	EDS File	29 KB

Figure 3



5.1.2 EDS file from the ifm website

1. Go to the [ifm global website](#) and select your country.
2. Enter the O3D3xx part number (e.g., O3D302) in the **search bar** and click **Search**.

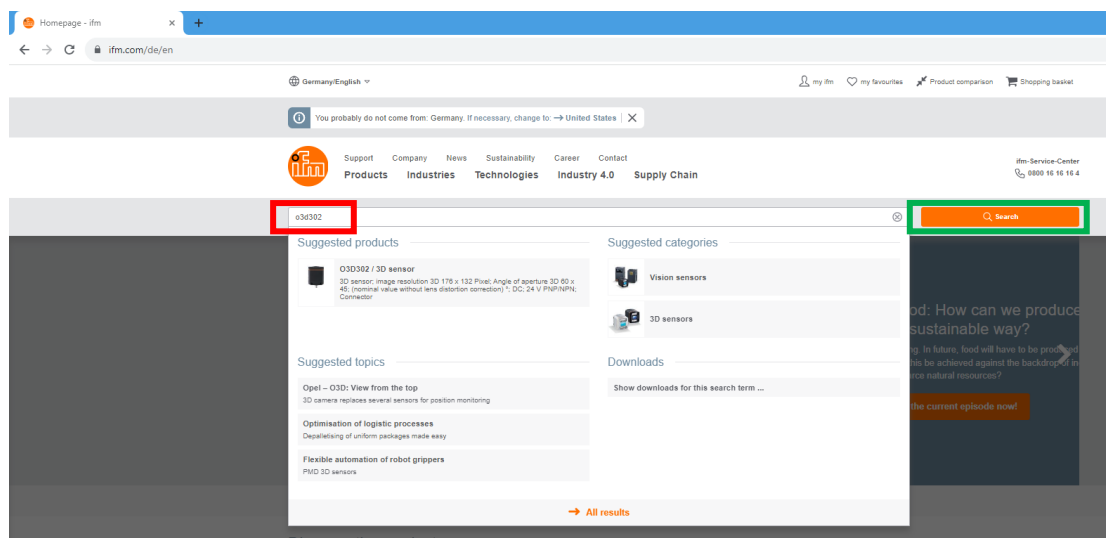


Figure 6

3. Click on the found **O3D3xx part number**.

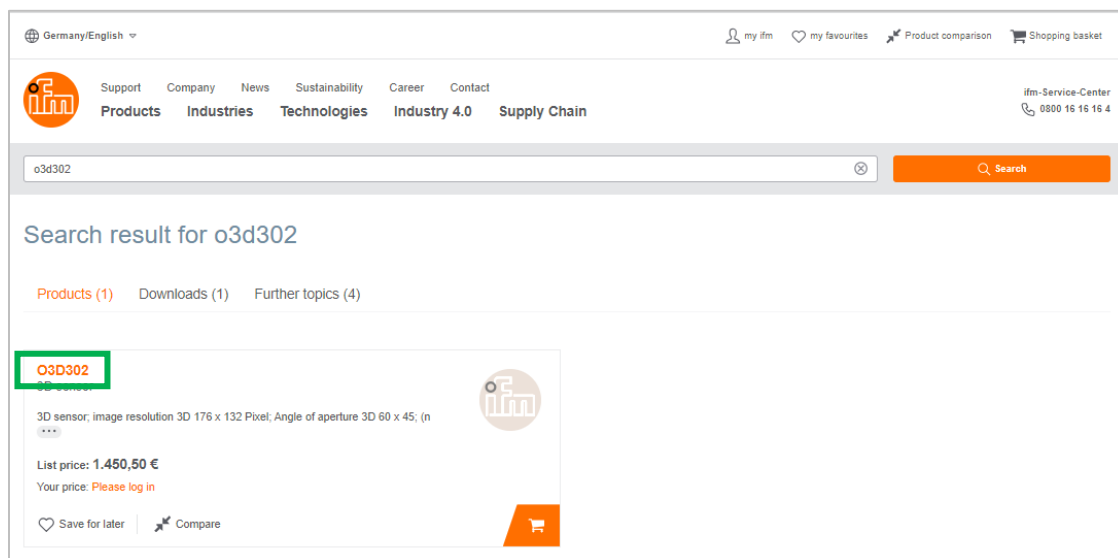


Figure 7



4. Click Downloads



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Enter search term.
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[Products](#) > [Image processing / Identification](#) > [Vision sensors](#) > [3D sensors](#) > **O3D302**





3D sensor

O3D302

O3DIRDKG/E1/GM/S/60

- Reliable volume determination for storage and conveyor technology
- Non-contact dimensioning of rectangular objects such as cardboard packages or parcels
- Reliable measurement of height, width and length to calculate strap length and volume
- Reliable detection of size, orientation and position of the objects for automated storage space planning
- Switching outputs and Ethernet process data interface

List price: **1.450,50 €**

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Accessories for O3D302



Connection cables with socket



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Success story



Behind the scenes of film's production of 2D and 3D vision systems



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Product characteristics

Image resolution 3D [px]	176 x 132
Area of analysis [m]	80° x 45° (horizontal and vertical beam deflection dimensions)

Figure 8



5. Find the **EDS file** in the downloads list and click **Download**.

Note: A “my ifm” account must be created and logged in before downloading.
Follow the prompts to create a “my ifm” account.

3D sensor
O3D302 O3D(RDKG)E1UGM/S/60

- Reliable volume determination for storage and conveyor technology
- Non-contact dimensioning of rectangular objects such as cardboard packages or parcels
- Reliable measurement of height, width and length to calculate strap length and volume
- Reliable detection of size, orientation and position of the objects for automated storage space planning
- Switching outputs and Ethernet process data interface

List price: **1,450.50 €** Without VAT ☐

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Accessories for O3D302

Technical details Accessories **Downloads** Further information

Startup package O3D3xx sensor

Startup package for O3D3xx sensor Contents: ifmVisionAssistant, firmware, VSMonitoring Tool, GSDML files, PLC example programs for Mitsubishi GX03UDE, Rockwell, BIP, AC14xx, URCap – robot pick/place application

Name	Description	File info	
ifmVisionAssistant Version 2.7.4	ifmVisionAssistant V2.7.4	zip(162.6 MB) SHA-256	Download
O3D3xx Firmware 1.71.10121	O3D3xx Firmware Version 1.71.10121	zip(111.8 MB) SHA-256	Download
EDS V2.2	EthernetIP EDS Data for O3D3xx	zip(0.1 MB) SHA-256	Download
GSDML V2.32	Profinet GSDML Data 64 Byte for O3D3xx Sensor	zip(0.1 MB) SHA-256	Download
Profinet GSDML file	456 bytes for O3D3xx sensor	zip(0.1 MB) SHA-256	Download

Figure 9

6. The EDS file is downloaded as a zip file to the Downloads folder.

Name	Date modified	Type	Size
ifm_O3D3xx_EthernetIP_EDS_File_V2.2.zip	1/4/2024 8:51 AM	Compressed (zip...)	3 KB

Figure 10

7. Unzip the folder to find the O3D3xx's eds file.

Name	Date modified	Type	Size
O3D3-EDS-V2.2.eds	5/24/2023 12:04 PM	EDS File	29 KB

Figure 11



5.2 EDS File installation

1. Add the EDS file to RS Linx by navigating to Tools->**EDS Hardware Installation Tool**.
2. Follow the prompts to add the EDS.

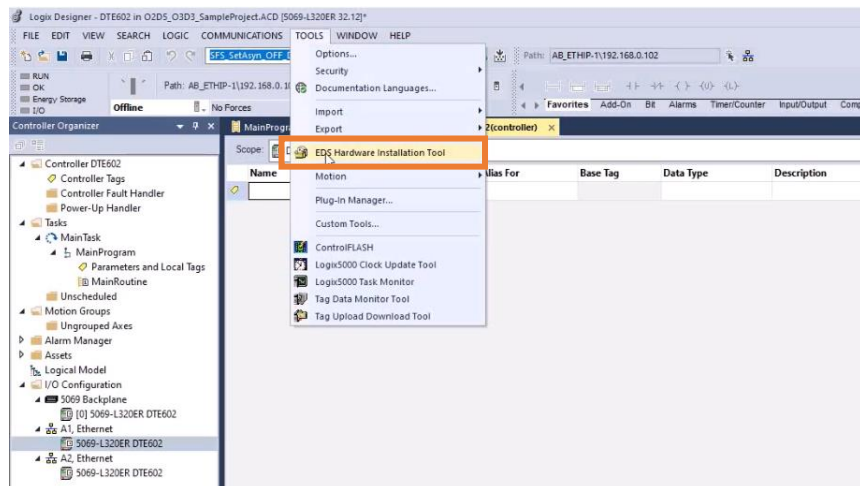


Figure 12

5.3 Add an O3D3xx Ethernet module to the PLC project

1. Right click on Ethernet in the I/O configuration folder and choose **New module**.

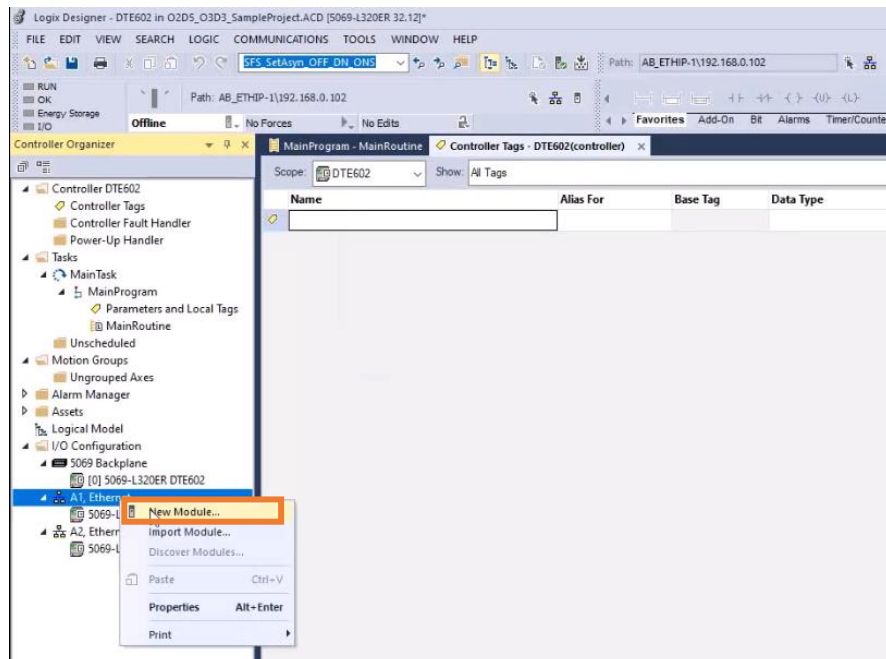


Figure 13



2. Search the article number for the sensor, i.e., **o3d**. Select the Catalog Number named **O3D3** and click **Create** button.

Select Module Type

Catalog Module Discovery Favorites

o3d Clear Filters Hide Filters

☒ Module Type Category Filters
☒ 20 - Comm-ER
☒ Analog
☒ CIP Motion Safety Track Section
☒ Communication
☒ Communications

☒ Module Type Vendor Filters
☒ Advanced Energy Industries, Inc.
☒ Bray International, Inc.
☒ Burkert Fluid Control Systems
☒ Cognex Corporation
☒ Dialight

Catalog Number	Description	Vendor	Category
O3D3	O3D3XX	ifm electronic gm...	Communications Adapter

1 of 787 Module Types Found Add to Favorites

☐ Close on Create Create Close Help

Figure 14

3. Provide a unique **Name** and **IP address** for the Ethernet module.
4. Click **Change** button to open the Module Definition Window.

New Module

General*
Connection
Module Info
Internet Protocol

General

Type: O3D3 O3D3XX
Vendor: ifm electronic gmbh
Parent: Local
Name: O3D3xx
Description:

Ethernet Address
☐ Private Network: 192.168.1.
☒ IP Address: 192 . 168 . 0 . 69
☐ Host Name:

Module Definition
Revision: 1.005
Electronic Keying: Compatible Module
Connections: Assembly object input data

Change ...

Status: Creating OK Cancel Help

Figure 15



5. In the Module Definition window, select **Exclusive Owner** in the name field and change the size type from **SINT** to **INT**.

Module Definition*

Revision: 1 005

Electronic Keying: Compatible Module

Connections:

Name	Remote Data	Size	Tag Suffix
Exclusive Owner	Input: Input_CP	101 225	1 03D3xx:I1
	Output: Output_CP	100 4	03D3xx:O1

OK Cancel Help

Figure 16

6. Verify the configuration of the O3D3xx sensor in the PLC project by navigating to the Controller Organizer. Double click on **Controller Tags** to find the **Input Array** of size 225 INT (450 bytes) and **Output Array** of size 4 INT (8 bytes) assigned to the O3D3xx's Ethernet module.

Logix Designer - DTE602 in O2D5_O3D3_SampleProject.ACD [5069-L320ER 35.11]*

File Edit View Search Logic Communications Tools Window Help

Path: Ethernet\192.168.0.102*

Controller Organizer

Controller Tags - DTE602(controller)

Scope: DTE602 Show: All Tags

Name	Alias For	Base Tag	Data Type	Description
03D3xx:I1				
03D3xx:I1.ConnectionFaulted			BOOL	
03D3xx:I1.Data			INT[225]	
03D3xx:O1				
03D3xx:O1.Data			INT[4]	

Figure 17



5.4 Requested Packet Interval (RPI) of the O3D3xx Ethernet module

The RPI time of the O3D3xx's Ethernet/IP module is the polling rate in which the PLC will update the input and output controller tags. The default setting is 200 ms. The possible setting range is 10.0-9999.9 ms.

1. To adjust the RPI time, right click on the **Ethernet/IP module** and select **Properties**.

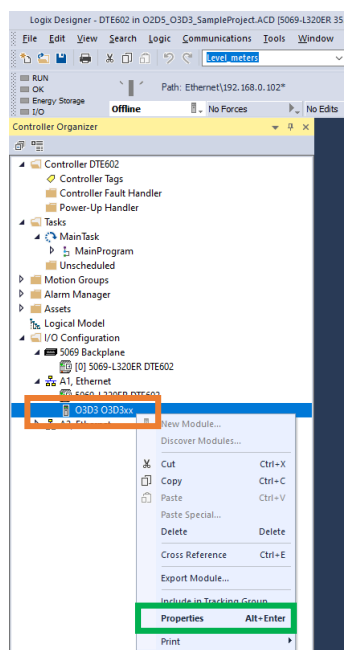


Figure 18

2. In the **Module Properties** window, select **Connection** and adjust the **RPI** field within the 10.0-9999.9 range. Click **OK** to accept the change.

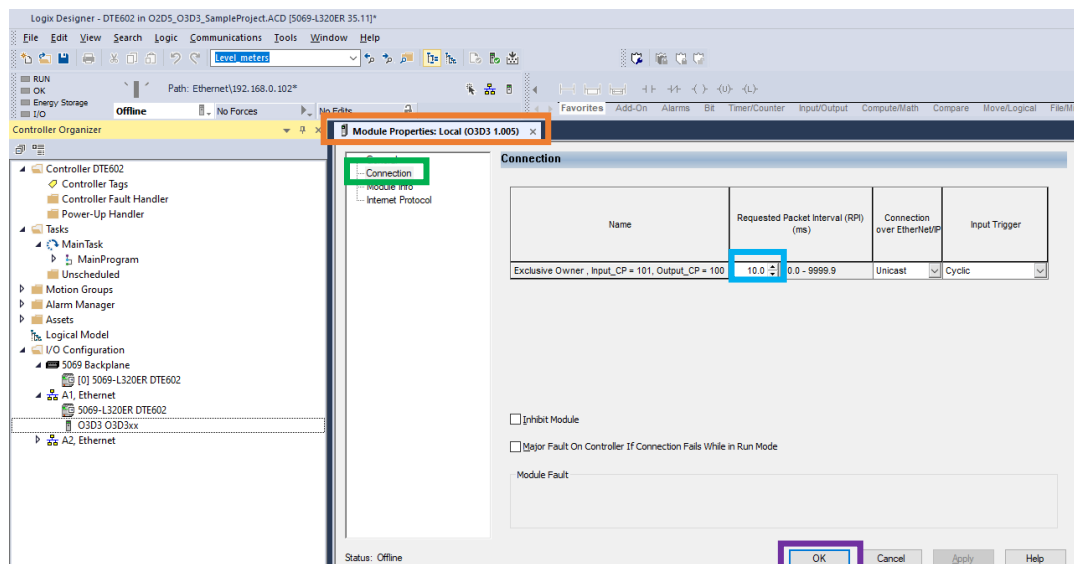


Figure 19



6 Integrate the ifm_O2D5_O3D3_Vision to the PLC project's logic

Section 6.1 and 6.2 show instructions while offline with the PLC. It is also possible to perform the importing methods while online.

6.1 Import the Add-On Instruction

1. In the Controller Organizer, expand the **Assets** folder. Right click Add-On Instruction and select **Import Add-On Instruction...**

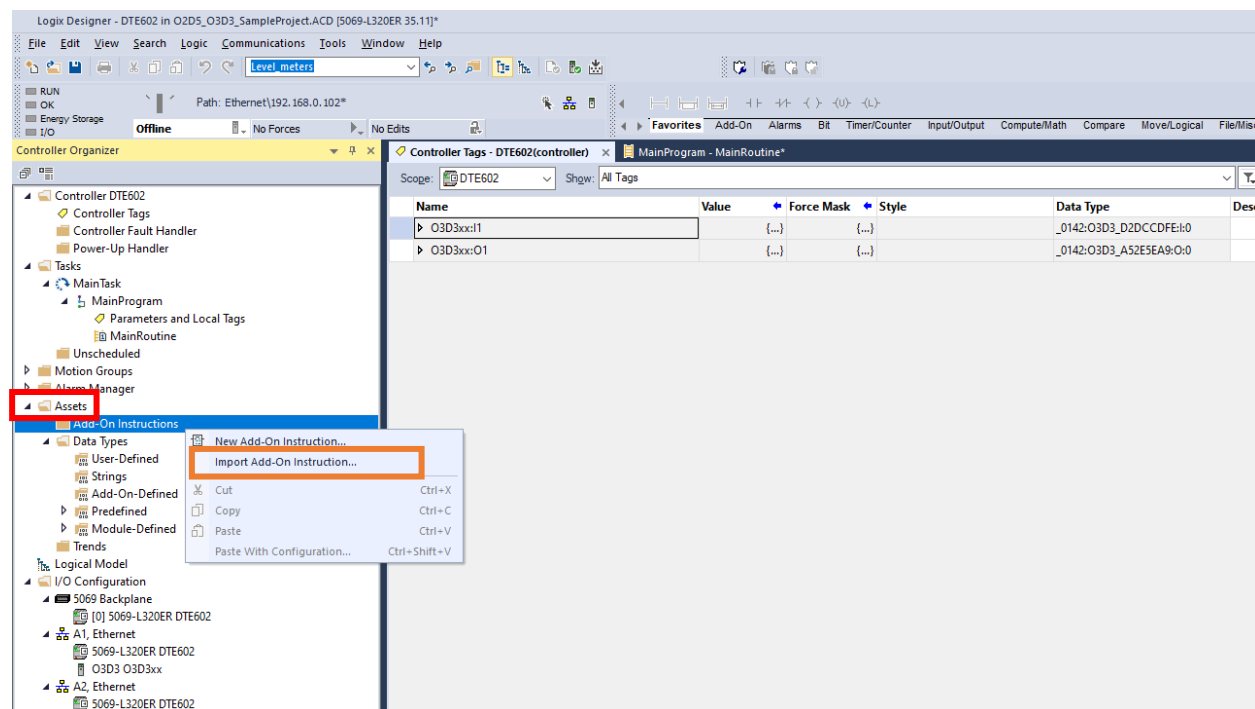


Figure 20



2. Select **ifm_O2D5_O3D3_Vision.L5X** from the folder named “2. Add_On Instruction” in the O3D3xx_EIP_AllenBradley_StartupPackage.

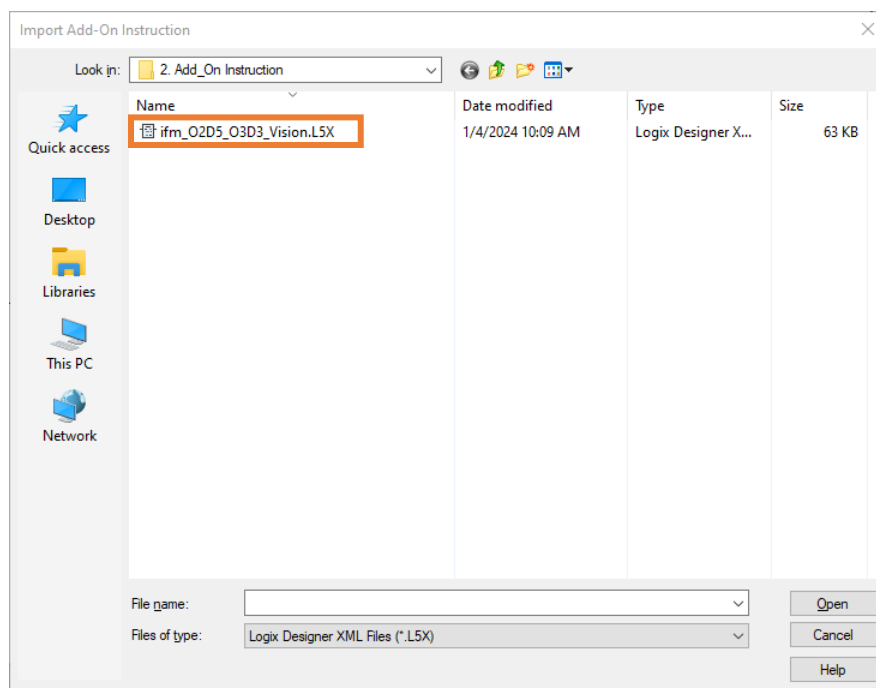


Figure 21

3. Click **OK** to Import.

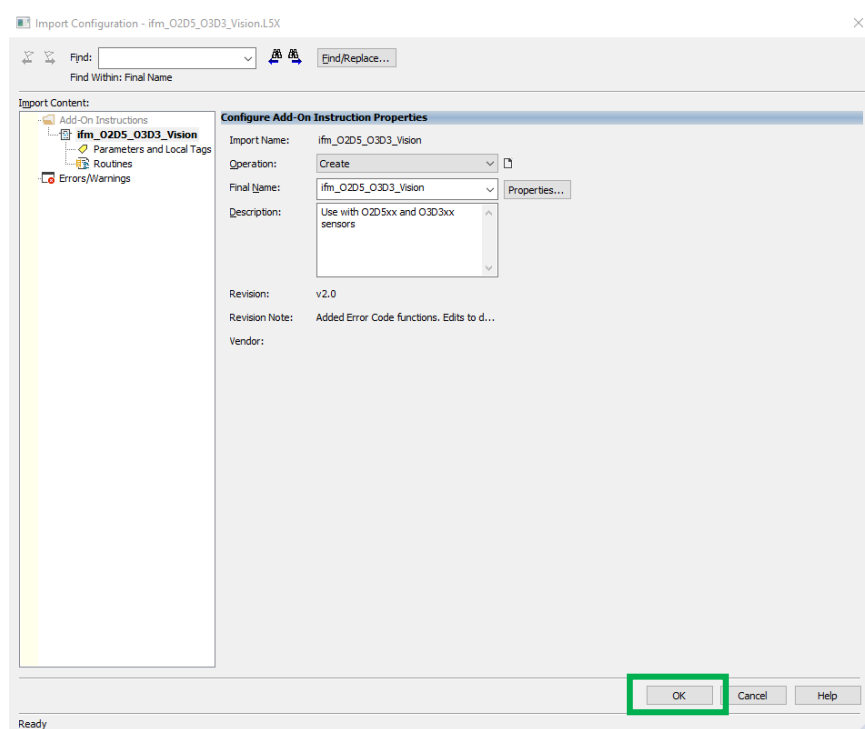


Figure 22



4. The **ifm_O2D5_O3D3_Vision** Add-On Instruction is displayed in the Add-On Instructions folder.

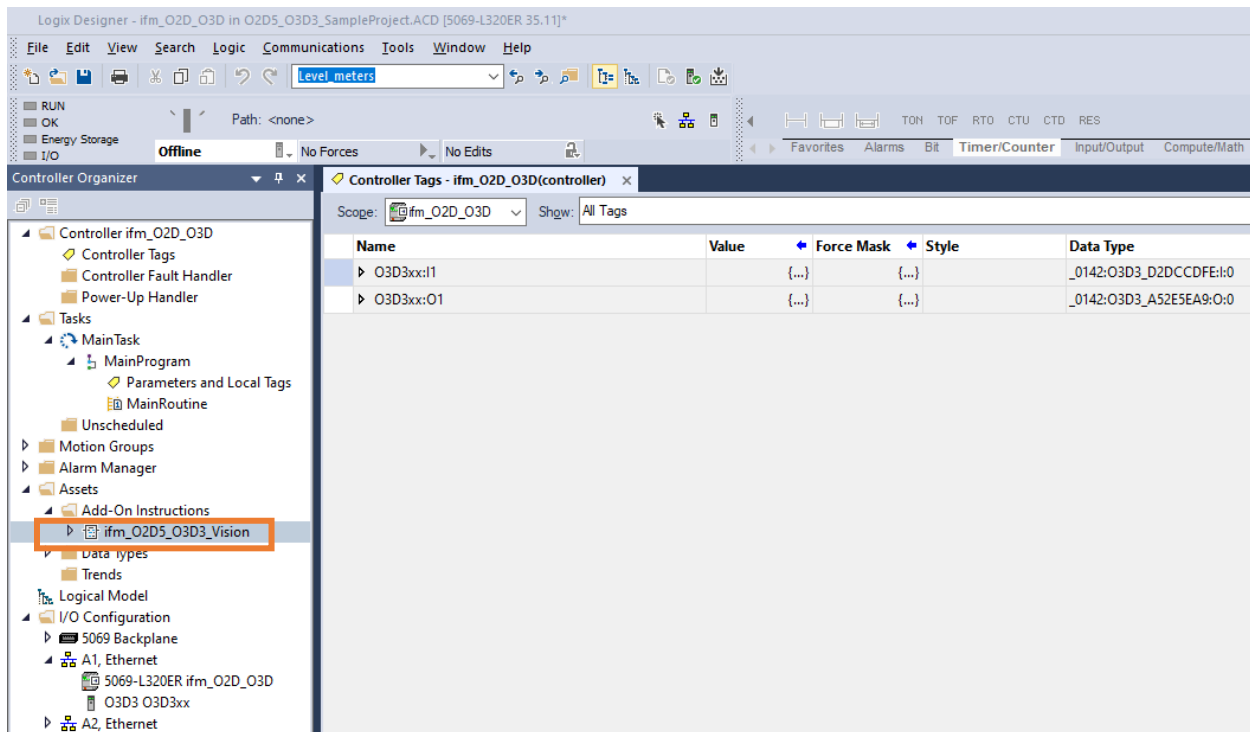


Figure 23



6.2 Integrate the ifm_O2D5_O3D3_Vision Add-On Instruction in a routine

1. Click and drag the **ifm_O2D5_O3D3_Vision** Add-On Instruction onto a rung of logic.

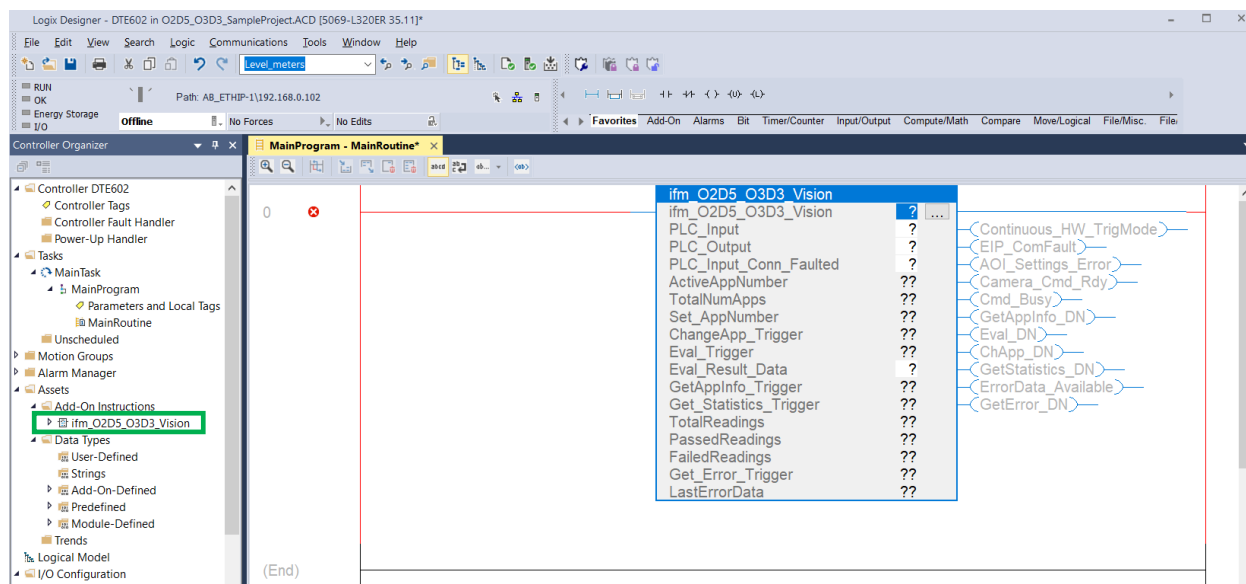


Figure 24



2. Define the Add-On Instruction tag name. Right click ? and click **New Tag**.

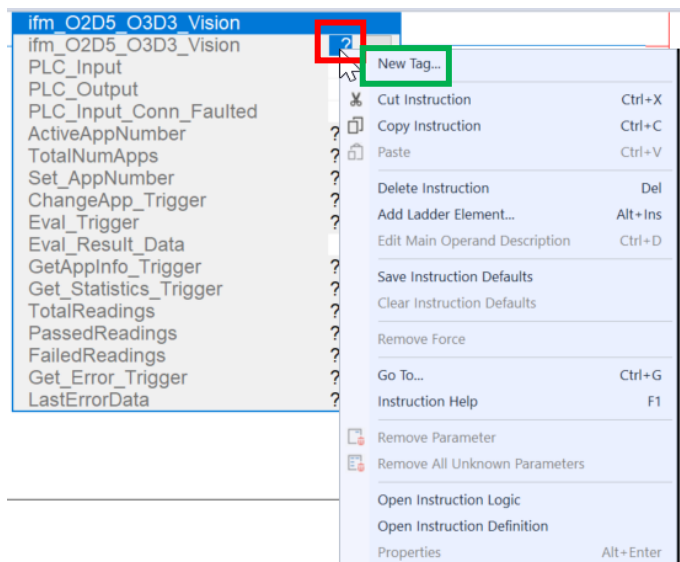


Figure 25

3. Create a **user defined name** and click **Create**.

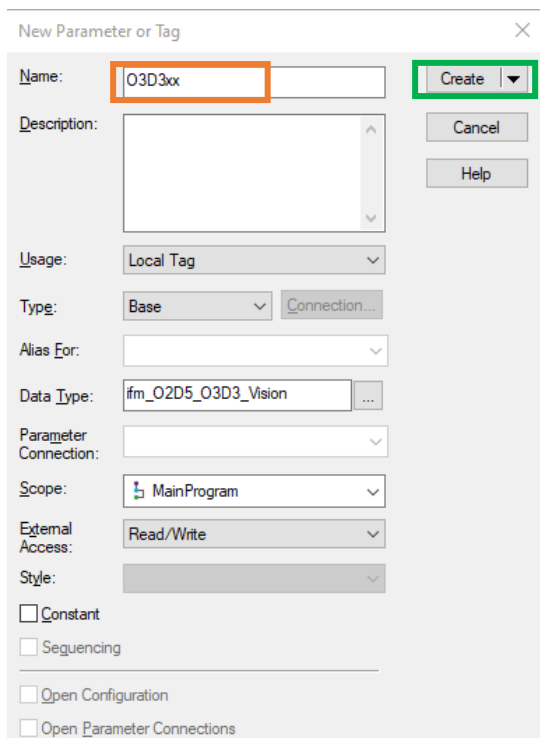


Figure 26



- Assign the PLC_Input and PLC_Output tags using the O3D3xx's producing **input data array** and consuming **output data array** from the Controller Tags.

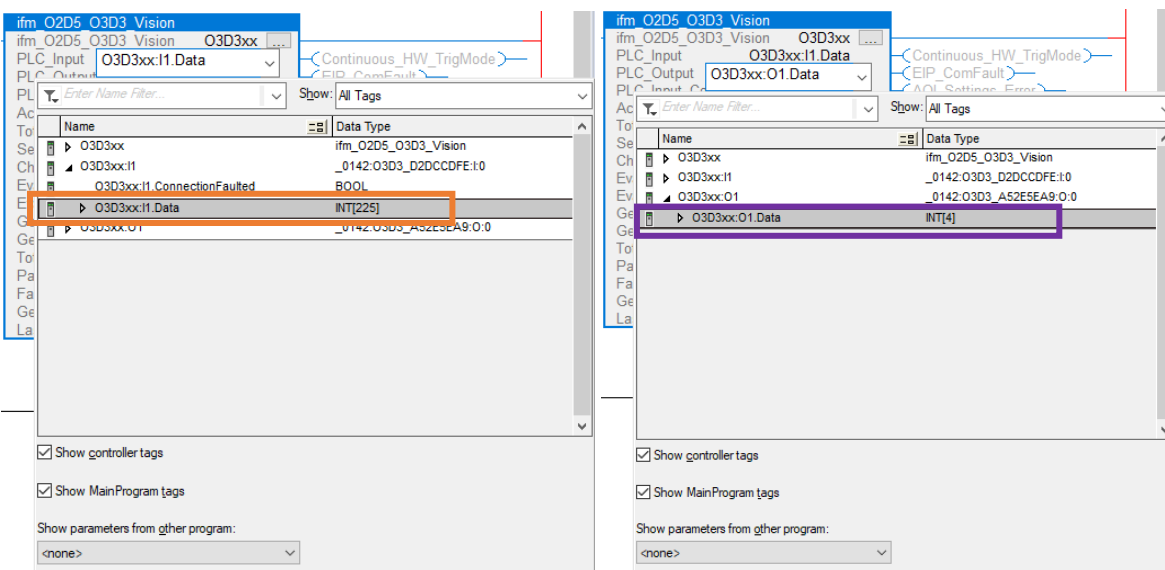


Figure 17

Figure 27

- Assign the **PLC_Input_Conn_Faulted** from the input Controller Tags.

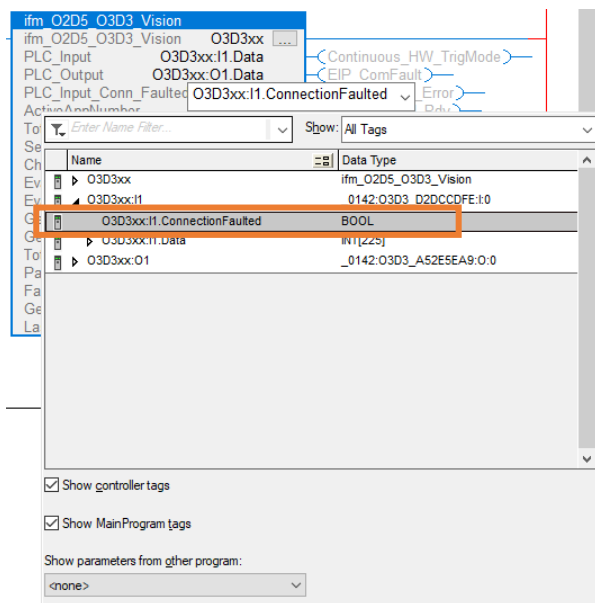


Figure 28



6. Right click ? and click **New Tag** to define the Add-On Instruction's Eval_Result_Data tag name.

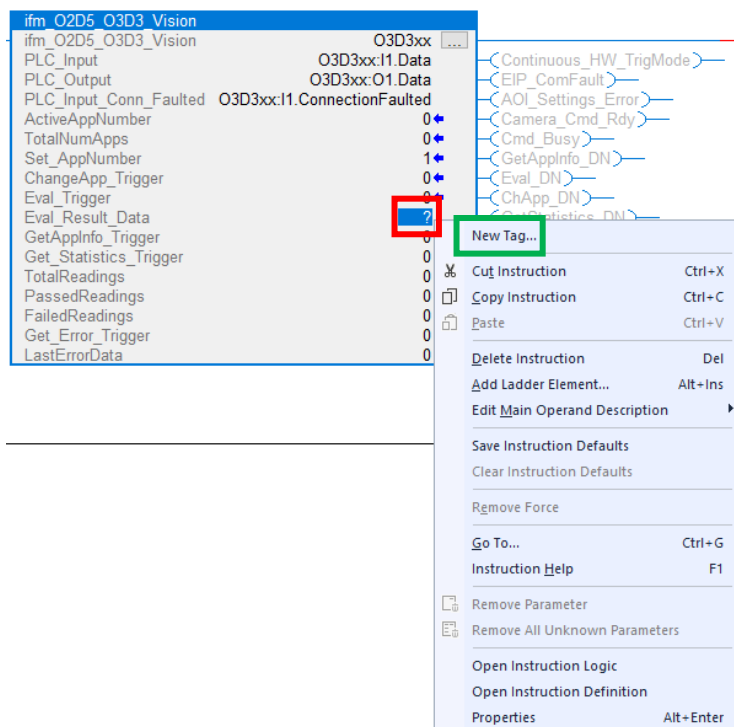


Figure 29

7. Create a **user defined name** and click **Create**.

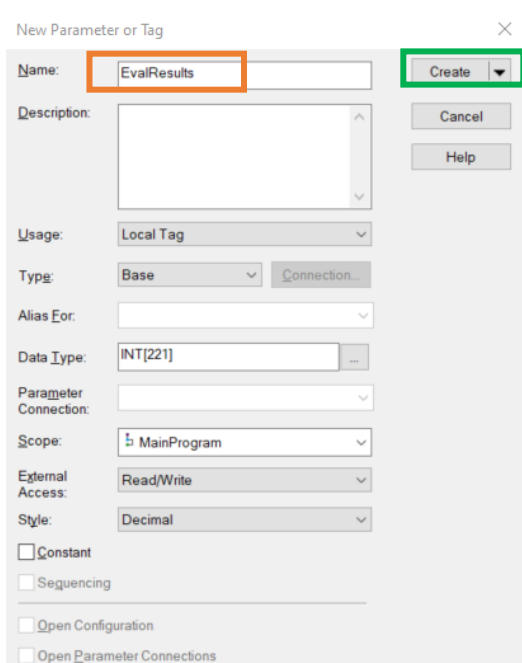


Figure 30



8. The rung is complete with no errors.

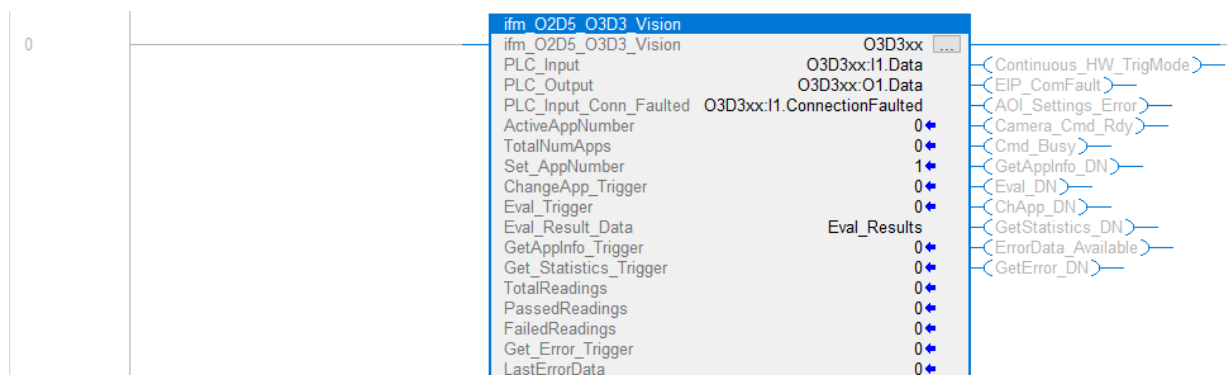


Figure 31

9. If the previous steps were made offline, the Add-On Instruction is ready to be downloaded to the PLC.
If the previous steps were made online, the pending edits are ready to be finalized.



7 Handling

7.1 ifm_O2D5_O3D3_Vision Add-On Instruction after an offline configuration

This section will explain the behavior of the ifm_O2D5_O3D3_Vision Add-On Instruction if the following conditions are met:

1. The Add-On Instruction is configured offline.
2. The PLC program is downloaded.
3. The O3D3xx sensor's Ethernet/IP communication is in a running state and the **AOI_Settings_Error** is false when the PLC is set to run mode.

If the offline configuration conditions apply, then the Add-On Instruction will execute a Get Application Information command on the first PLC scan.

First PLC Scan executes GetAppInfo Command:

Cmd_Busy =TRUE

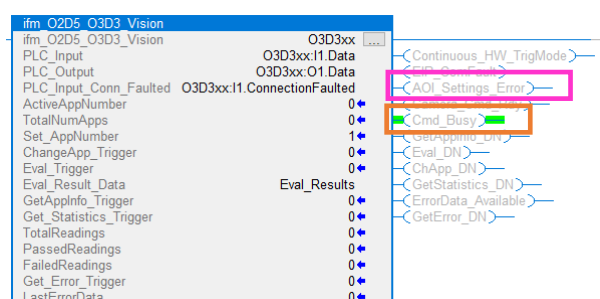


Figure 32

GetAppInfo processing complete:

Cmd_Busy =FALSE
GetAppInfo_DN =TRUE
ActiveAppNumber =Value range 1..32
TotalNumApps =Value range 1..32
Continuous_HW_TriggerMode =TRUE or FALSE (See Note)
Camera_Cmd_Rdy =TRUE

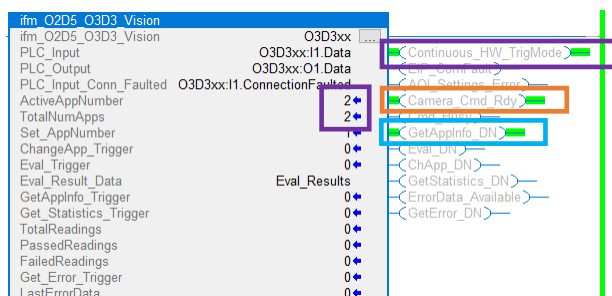


Figure 33

Note: The Continuous_HW_TriggerMode output is set according to the O3D3xx's trigger configuration. See [section 7.3](#) for an explanation of Continuous_HW_TriggerMode behavior.



7.2 ifm_O2D5_O3D3_Vision Add-On Instruction after an online configuration

This section will explain the behavior of the ifm_O2D5_O3D3_Vision Add-On Instruction if one of the following conditions are met:

1. The Add-On Instruction is added to a rung of logic while making an online edit.
2. The Add-On Instruction was configured while offline, but at the time of the first PLC scan the AOI_Settings_Error output was true.

In both of the above cases, the Add-On Instruction's first scan logic does not execute and therefore the Get Application Information command is not performed by the Add-On Instruction. To update the ActiveAppNumber and TotalNumApps outputs, the user must execute the command.

Online edits are finalized and no command is processing:

Camera_Cmd_Rdy =TRUE
ActiveAppNumber =0
TotalNumApps =0
Continuous_HW_Trig_Mode =FALSE

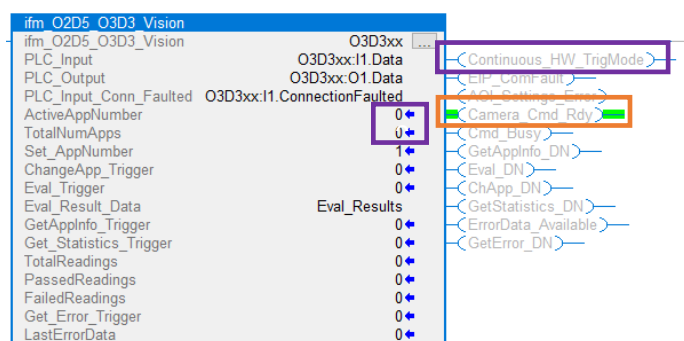


Figure 34



1. The user executes the Get Application Info command:

Command Processing:

GetApplInfo_Trigger =TRUE

Cmd_Busy =TRUE

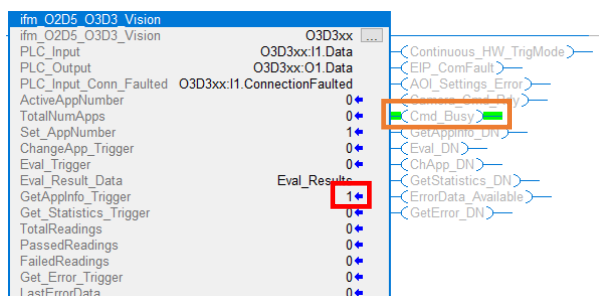


Figure 35

GetApplInfo processing complete:

GetApplInfo_Trigger =TRUE or FALSE

Cmd_Busy =FALSE

GetApplInfo_DN =TRUE

ActiveAppNumber =Value range 1..32

TotalNumApps =Value range 1..32

Continuous_HW_TriggerMode =TRUE or FALSE (See Note)

Camera_Cmd_Rdy =TRUE or FALSE (See Note)

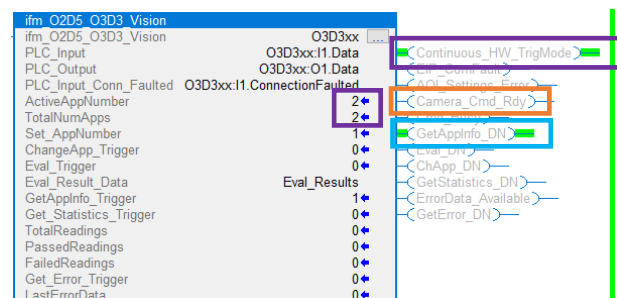


Figure 36

Note: The Continuous_HW_TriggerMode is set according to the O3D3xx's trigger configuration.

See [section 7.3](#) for an explanation of Continuous_HW_TriggerMode behavior.

After the command is processed the Camera_Cmd_Rdy is set according to the state of the GetApplInfo_Trigger. See [section 7.4.2](#) for an explanation of Camera_Cmd_Rdy behavior.



7.3 Trigger Modes: Continuous_HW_Trigger

After the Get Application Info command has completed, the ifm_O2D5_O3D3_Vision Add-On Instruction monitors for any triggering of evaluations.

Process Interface Trigger applications:

If the configuration of the O3D3xx sensor is set to Process Interface:
Continuous_HW_TrigMode output is set FALSE.

Continuous or Hard-Wired Trigger applications:

If the O3D3xx's trigger configuration is set to Continuous:
Continuous_HW_TrigMode output is set TRUE after the O3D3xx is set to monitor mode and the first evaluation is processed.

If the O3D3xx trigger configuration is set to Positive Edge, Negative Edge or Positive and Negative Edge:
Continuous_HW_TrigMode output is set TRUE after the first edge-controlled evaluation is processed.

The Continuous_HW_TriggerMode output will hold TRUE until:

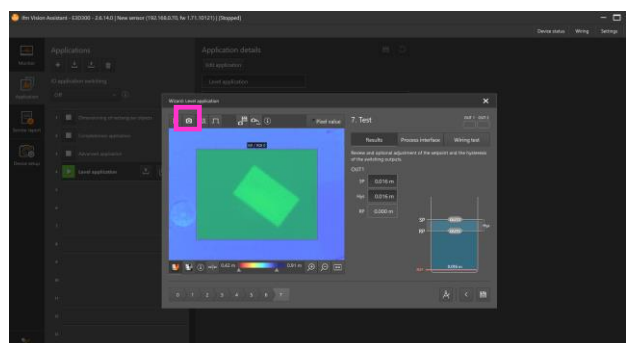
1. The Change Application command is processed by the rising edge of ChApp_Trigger.
2. The Get Application Information command is processed by the rising edge of GetAppInfo_Trigger.
3. An Ethernet/IP communication fault.
4. The sensor is rebooted.

Vision Assistant Test operations:

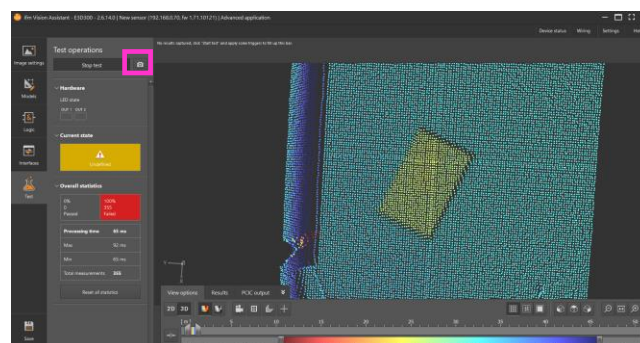
The last step in the Vision Assistant configuration of the O3D3xx is Test operations. In Test operations the O3D3xx sensor behaves as it would in Monitor mode to the PLC.

If the trigger configuration is set to Process Interface and the Vision Assistant's **evaluation button** is utilized by the user, the Add-On Instruction interprets the evaluation's source to be from a continuous or hard-wired trigger and the Continuous_HW_Trig output will be set TRUE.

When the O3D3xx is set back to monitor mode, the Get Application Info command must be executed to refresh the Continuous_HW_TrigMode output so the Eval_Trigger may be used. See [section 7.4.8](#) for an explanation of the Get Application Info command.



Test Mode when using configuration Templates



Test Mode when using User-defined mode

Figure 37



7.4 Command Processing

7.4.1 Command Processing Sequence

If the Camera_Cmd_Rdy is true, the O3D3xx is ready to process a command. A command is started with the rising edge of any of the ifm_O2D5_O3D3_Vision Add-On Instruction's trigger inputs (tags ending in "Trigger.")

O3D3xx ready for command:

Camera_Cmd_Rdy	=TRUE
xxxx_Trigger	=FALSE
Cmd_Busy	=FALSE
xxxx_DN	=TRUE or FALSE

Command started:

Camera_Cmd_Rdy	=FALSE
xxxx_Trigger	=TRUE
Cmd_Busy	=TRUE
xxxx_DN	=FALSE

Command in process:

Camera_Cmd_Rdy	=FALSE
xxxx_Trigger	=TRUE or FALSE
Cmd_Busy	=TRUE
xxxx_DN	=FALSE

Command processing complete:

Camera_Cmd_Rdy	=FALSE / TRUE (See Note)
xxxx_Trigger	=TRUE / FALSE (See Note)
Cmd_Busy	=FALSE
xxxx_DN	=TRUE

Note: xxxx =Eval, ChApp, GetAppInfo, GetStatistics, GetError

If xxxx_Trigger is true when the command is finished processing the Camera_Cmd_Rdy output remain false. The Camera_Cmd_Rdy output will turn true when the xxxx_Trigger is set false and the O3D3xx is ready for a command.

If the xxxx_Trigger is false before the command is finished processing the Camera_Cmd_Rdy output will turn true when the O3D3xx is ready for a command.



7.4.2 Camera Ready for a Command: Camera_Cmd_Rdy

The O3D3xx sensor is ready to process a command:

Camera_Cmd_Rdy =TRUE

A command is started on a rising edge of any trigger (tags ending in "Trigger"):

Command in process:

Camera_Cmd_Rdy =FALSE

Command processing is completed when the command's done output (tags ending in "DN") is true.

Command processing complete and command trigger is false:

xxxx_DN =TRUE

xxxx_Trigger =FALSE

Camera_Cmd_Rdy =TRUE

Command processing complete and command trigger is true:

xxxx_DN =TRUE

xxxx_Trigger =TRUE

Camera_Cmd_Rdy =FALSE

The O3D3xx loses Ethernet communication or is rebooted:

Camera_Cmd_Rdy =FALSE

7.4.3 Command Busy: Cmd_Busy

Command in process:

Cmd_Busy =TRUE

Command processing complete:

Cmd_Busy =FALSE

The O3D3xx loses Ethernet communication or is rebooted:

Cmd_Busy =FALSE



7.4.4 Done Outputs – GetAppInfo_DN, Eval_DN, ChApp_DN, GetStatistics_DN or GetError_DN

There are five done outputs from the ifm_O2D5_O3D3_Vision Add-On Instruction.

1. GetAppInfo_DN
2. Eval_DN
3. ChApp_DN
4. GetStatistic_DN
5. GetError_DN

Initial setting of all xxxx_DN outputs: =FALSE

Command processing complete:

xxxx_DN =TRUE

The following conditions will reset the one or more done outputs:

1. If any done output is true, it will be set false on the rising edge of its associated trigger.
2. The rising edge of the ChApp_Trigger will set the ChApp_DN, GetAppInfo_DN and GetStatistic_DN outputs false.
3. If the O3D3xx loses Ethernet communication or is rebooted, all done outputs are set false.

7.4.5 Triggers

There are five triggers available in ifm_O2D5_O3D3_Vision Add-On Instruction:

1. GetAppInfo_Trigger
2. Eval_Trigger
3. ChangeApp_Trigger
4. Get_Statistics_Trigger
5. Get_Error_Trigger

A rising edge of a trigger starts the processing of its associated command if the Camera_Cmd_Rdy output is true.

A trigger input may be reset to false before the command is done processing; i.e., it can be used with a ONS instruction.

If multiple triggers are set true in the same PLC scan no commands are executed.

Multiple triggers set true:

1. If a trigger is true, no command is processing and another trigger is set true, then no command is executed.
2. If a command is in process and one or more triggers are set true, the command in process will finish and no new commands will be executed.

In both of the above cases, all triggers must be set false before the Add-On Instruction will execute another command.



7.4.6 Process Interface (Ethernet/IP) Trigger: Eval_Trigger

The rising edge of the Eval_Trigger sends a process interface (Ethernet/IP) trigger command to the O3D3xx sensor. The sensor will perform an evaluation as configured by the Vision Assistant.

When the Evaluation command is complete:

Eval_DN	=TRUE
Eval_Result	=data array is updated

7.4.7 Evaluation Results: Eval_Results

The evaluation results are updated after a process interface (Ethernet/IP) trigger command has completed processing. The Eval_Results is a data array of integers with a size of 221. The Eval_Result data array mapping is configured by the Process Interface section in the Vision Assistant.

7.4.8 Get Application Information: GetAppInfo_Trigger

The rising edge of the GetAppInfo_Trigger processes the get application information command. The command will set the ActiveAppNumber and TotalNumApps values.

The command is automatically executed by the Add-On Instruction if the following conditions are met:

1. On a first scan of the PLC. See [section 7.1](#) for a description of when the first scan is applicable.
2. After a change application command is completed, the Add-On Instruction will execute a GetAppInfo command.
3. The O3D3xx loses Ethernet communication or is rebooted.

The command should be executed by the user if any of the following changes are made using the Vision Assistant and the O3D3xx is set back to Monitor Mode:

1. Addition of an application.
2. A change to the active application.
3. The Trigger mode of an application is changed.
4. When in the Test operations of an application in which the trigger configuration is set to Process Interface and the evaluation button is used to trigger the O3D3xx. See [section 7.3](#) for more detail.

7.4.9 Set Application Number: SetAppNumber

The O3D3xx's application numbers are assigned by the Vision Assistant. Up to 32 applications may be saved on the O3D3xx sensor. The ifm_O2D5_O3D3_Vision Add-On Instruction may be used to switch active applications. The SetAppNumber is the application number to be made active after a change application command is executed.



7.4.10 Change Applications: ChangeApp_Trigger

The rising edge of the ChangeApp_Trigger processes the Change Application command. The command will change the O3D3xx's application to the application set by the SetAppNumber value.

Command processing complete:

ChApp_DN	=TRUE
ActiveAppNumber	=Value range 1..32
TotaNumApps	=Value range 1..32
Continuous_HW_TrigMode	=TRUE or FALSE

7.4.11 Get Statistics: Get_Statistic_Trigger

The rising edge of the Get_Statistics_Trigger processes the get statistics command.

Command processing complete:

Get_Statistics_DN	=TRUE
TotalReadings	=Value of total readings (Passed + Failed)
PassedReadings	=Value of passed readings
FailedReadings	=Value of failed readings

The TotalReadings, PassedReadings and FailedReadings values:

1. Are reset by the Vision Assistant.
2. Are set to 0, when the Add-On Instruction is used to change changing applications. When the Get Statistics command is executed the values are updated to the current settings as held by the O3D3xx.

7.4.12 Error Processing: Get_Error_Trigger

If the O3D3xx processes an error:

ErrorData_Available	=TRUE
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Get Last Error Command processing:

Get_Error_Trigger	=TRUE
Cmd_Busy	=TRUE
GetError_DN	=FALSE

Get Last Error Command is complete:

Get_Error_Trigger	=TRUE or FALSE
Cmd_Busy	=FALSE
GetError_DN	=TRUE
ErrorData_Available	=FALSE
LastErrorData	=Error Code ID

See table in [section 9](#) for a list of Error Code ID's and their meanings

Reset LastErrorData =0 (Optional):

Execute the Get_Error_Trigger when the ErrorData_Available is false. When the command has completed the LastErrorData's value will be set to 0.



7.5 Communication Fault: EIP_ComFault

If the Ethernet/IP communication is interrupted:

EIP_ComFault =TRUE

When the communication is reestablished:

EIP_ComFault =FALSE

7.6 User input errors: AOI_Settings_Error

If the AOI_Settings_Error is true, then at least one user input of the ifm_O2D5_O3D3_Vision Add-On Instruction is out of range or incorrectly set. When the user corrects the error, the AOI_Settings_Error will be false.

The following cases will cause an AOI_Setting_Error to be true:

1. The SetAppNum = ActiveAppNumber and the user sets the ChangeAppTrigger true.
2. GetApp_DN is true and the SetAppNum>TotalNumApps.
3. SetAppNum<=0. A negative value and zero are invalid application numbers for the O3D3xx.
The AOI_Settings_Error will be set false if GetApp_DN is false and SetAppNum >0 and <=32.
4. SetAppNum>32. The O3D3xx's maximum number of memory locations is 32.
5. The Eval_Trigger is true while the Continuous_HW_TrigMode output is true.



8 ifm_O2D5_O3D3_Vision Parameter Data Types and Descriptions

Name	Usage	Type	Description
PLC_Input	INT[225]	InOut Parameter	Input array from sensor
PLC_Output	INT[4]	InOut Parameter	Output array to sensor
Eval_Result_Data	INT[221]	InOut Parameter	Evaluation result data as defined in Vision Assistant
Get_Error_Trigger	DINT	Output	Last error code data retrieved by the Get_Error_Trigger
Set_AppNumber	SINT	Input	Number of applications to be set after ChangeApp_Trigger is set
ChangeApp_Trigger	BOOL	Input	Activate change application
Eval_Trigger	BOOL	Input	Enable trigger
GetAppInfo_Trigger	BOOL	Input	User controlled trigger to get the active application number
Get_Statistics_Trigger	BOOL	Input	User controlled trigger to get the number of total, passed and failed readings
EIP_Com_Faulted	BOOL	Output	Connection status of the EIP module
ActiveAppNumber	INT	Output	Active application number
TotalNumApps	DINT	Output	Total number of applications on the O3D3xx
TotalReadings	DINT	Output	Total number of evaluations since last reset
PassedReadings	DINT	Output	Number of passed readings since last reset
FailedReadings	DINT	Output	Number of failed readings since last reset
LastErrorData	DINT	Output	Last error code data retrieved by the Get_Error_Trigger
Continuous_HW_TrigMode	BOOL	Output	Sensor is continuously triggered as defined in the Vision Assistant
AOI_Settings_Error	BOOL	Output	At least one user input in the Add-On Instruction is out of range
O2x5xx_Cmd_Rdy	BOOL	Output	The O2x5xx is ready to process any Add-On Instruction trigger command
Cmd_Busy	BOOL	Output	A command is processing
GetAppInfo_DN	BOOL	Output	Rising edge indicates the application info (number of sensor applications and active application is refreshed)
Eval_DN	BOOL	Output	Rising edge indicates evaluation is complete after user has set the Eval_Trigger. Result data is available in the Eval_Result_Data
ChApp_DN	BOOL	Output	Rising edge indicates the changing of application is complete



Name	Usage	Type	Description
GetStatistics_DN	BOOL	Output	Rising edge indicates the total, passed and failed readings is refreshed
ErrorData_Available	BOOL	Output	An error has occurred in the O2x5xx. Use GetError_Trigger to retrieve the error code
GetError_DN	BOOL	Output	Rising edge indicates error code is present in the error codes buffer

Figure 38

9 Error Codes

Error Code ID	Description
100000001	Maximum number of connections exceeded
100000002	Internal failure during a D-Bus call
100000003	Unspecified internal error
100000004	Generic invalid parameter
100000005	Invalid command
100001000	Application configuration does not allow Process interface trigger
100001001	Video mode does not allow Process interface trigger
100001002	There is no application configured
100001003	Invalid image id in I? command
100001004	Invalid pin id in o/O? command
100001005	Invalid pin configuration in o/O? command
100001006	Invalid conversion type selected
100001007	No trigger was run yet
100001008	Missed decoded frame
100001009	No more segments left
100001010	Command bit 4 not reset to 0
110001001	Boot timeout
110001002	Fatal software error
110001003	Unknown hardware
100001004	Invalid pin id in o/O? command
100001005	Invalid pin configuration in o/O? command
110001006	Trigger overrun
110001007	Ethernet configuration was changed. The socket is going to be closed.
110002000	Short circuit on Ready for Trigger
110002001	Short circuit on OUT1
110002002	Short circuit on OUT2
110002003	Reverse feeding
110003000	Vled overvoltage
110003001	Vled undervoltage
110003002	Vmod overvoltage



Error Code ID	Description
110003003	Vmod undervoltage
110003004	Mainboard overvoltage
110003005	Mainboard undervoltage
110003006	Supply overvoltage
110003007	Supply undervoltage
110003008	VFEMon alarm
110003009	PMIC supply alarm
110004000	Illumination overtemperature
120000001	NTP server not available
120000002	other NTP error

Figure 39